* Can we write java code / can we execute java application without the main() method?

⇒ Yes it was possible to execute java code using {Static Initializer Blocks} instead of main method but only till Java version 1.6.
From version 1.7 main method is mandatory.

```
static {
    • env
    • ser
}
```

requirements.txt {docker, nginx}

* What is Constructor Chaining?

⇒ Constructor Chaining is the process of calling one constructor from another constructor of the same class using this() to reuse code & avoid duplication.

* It helps:
  * Code reuse
  * Maintainability
  * Clean initialization logic

4 Pillars of OOP :→

① Encapsulation :→ Wrapping / Binding the data members (vars, methods) inside a block of code (class) so that it is not accidentally modified, is called data encapsulation.

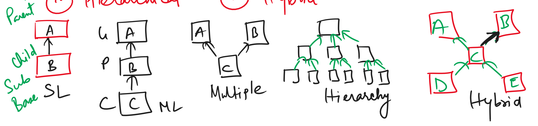
In Java, we use the "private" access modifier to achieve encapsulation.
To access the data members outside the class, we use two special "public" methods called "setters" & "getters". (return type)
(No return type)

* Inheritance: The property by which a child class can inherit data from a parent class is called inheritance. (Top Down Architecture)

Benefits:
* Reduces code duplication
* Increases code reusability
* Increases efficiency

Types of inheritances :→

① Single ② Multi ③ Multiple
④ Hierarchical ⑤ Hybrid



Grand Parent
Tata Nexon ①

Parent
Ford Endeavour ②

⇒ Child
XUV 700 (SN)

(*) {Access increases level by level}

(*) The lowest level gets access to all data from the top levels.

[Diamond Problem]

Class Animal
makeSound() → any sound!

Class Cat
makeSound() → meow!

Class Dog
makeSound()
→ woof!

Class Cog
makeSound(); (XXX)

Ambiguity

Can a class extend multiple classes in Java? (Interfaces)
(NO)

* Java doesn't support multiple inheritance or hybrid inheritance directly because of the "Diamond Problem" scenario.
* To achieve multiple inheritance we use a new entity called "interface".
* An interface is a completely abstract class.